AI Research Engineer (Multi-Modal & Vision)

This position is listed on behalf of a partner company, who manages all applications and next steps.
Our partner is looking for an AI Research Engineer (Multi-Modal & Vision) based in Canada.
This is an exciting opportunity for a research-focused AI engineer to contribute to the development of advanced multimodal systems that combine vision and language capabilities. The role covers the full AI model lifecycle, from dataset creation and training pipeline development to model evaluation, optimization, and deployment. Working within a highly skilled and collaborative team, you will help build scalable AI solutions designed for real-world production environments. The position offers significant ownership, direct impact on cutting-edge research initiatives, and the opportunity to apply state-of-the-art techniques to solve complex challenges. Ideal candidates are passionate about advancing multimodal AI while maintaining a strong engineering mindset focused on measurable outcomes and practical deployment.

Accountabilities
Conduct end-to-end research and development of vision-language models, including training, evaluation, optimization, and deployment activities.
Design and implement advanced post-training methodologies such as supervised fine-tuning, knowledge distillation, and reinforcement learning from human feedback.
Build, curate, filter, and maintain high-quality multimodal datasets tailored to domain-specific applications.
Improve model efficiency and scalability through optimization, compression, and adaptation techniques suitable for resource-constrained environments.
Develop benchmarking systems and evaluation frameworks to assess model quality, robustness, and real-world performance.
Build and maintain distributed training workflows across GPU infrastructure while identifying and resolving performance bottlenecks.
Contribute to open-source AI ecosystems by leveraging and enhancing models, datasets, and development tools.
Monitor emerging research in multimodal learning and vision-language systems, translating relevant advancements into practical improvements.
Collaborate on research publications and contribute to scientific advancements through conference or journal submissions when appropriate.
Requirements

Bachelor's degree in Computer Science, Machine Learning, Artificial Intelligence, or a related field; Master's or PhD preferred.
Strong hands-on experience working with multimodal AI systems, particularly vision-language models.
Proven expertise in supervised fine-tuning, knowledge distillation, reinforcement learning from feedback, and other post-training optimization techniques.
Experience with parameter-efficient fine-tuning approaches and distributed training frameworks.
Demonstrated success improving model performance on industry-standard benchmarks or production use cases.
Strong understanding of model optimization techniques for deployment in resource-constrained environments.
Experience building scalable machine learning pipelines and training workflows on GPU infrastructure.
Proven contributions to open-source multimodal AI projects through platforms such as GitHub or Hugging Face.
Research background supported by publications in leading AI conferences or journals is highly desirable.
Strong analytical thinking, problem-solving skills, and the ability to balance research innovation with production-oriented engineering practices.
Excellent communication skills and the ability to collaborate effectively within distributed, cross-functional teams.
Benefits
Competitive salary package aligned with experience and expertise.
Opportunity to work on cutting-edge multimodal AI and vision-language research projects.
Fully remote work environment with global collaboration opportunities.
Exposure to large-scale AI infrastructure and advanced machine learning technologies.
High degree of autonomy, ownership, and impact on product and research outcomes.
Collaboration with experienced researchers, engineers, and AI specialists.
Professional growth opportunities through research, innovation, and publication support.
Flexible working arrangements that support work-life balance.
Dynamic, fast-paced environment focused on innovation and continuous learning.